balanced_leverage

equity strategy | family: equity-neutralized | status: validated | derived from `sector_balanced`

What this strategy does

Sector Balanced, allowed to hold up to 2.5x the account in positions

Why it should work. Market-neutral earnings drift with enough gross exposure to matter; the neutrality is the hedge.

Design notes (from the strategy's source file).

Balanced Leverage -- Sector Balanced, but allowed to hold bigger positions (2.5x the account).

Sector Balanced's weights, but with the 1.5x cap lifted to 2.5x (the point where it stops binding) and sizing retuned. Because sector+size neutralization already prevents concentration, the cap was protecting nothing: return rises 4.1% -> 6.5% with the Sharpe holding at ~1.3. The pure-alpha reference strategy for the repo.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

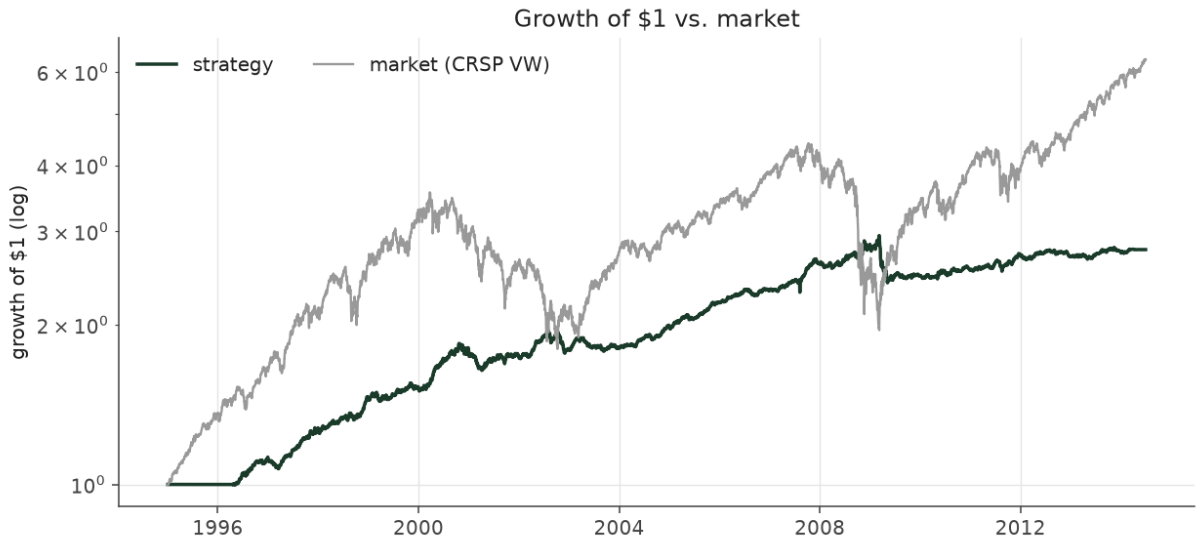
metric	balanced leverage	market (CRSP VW)
Annualized return	5.38%	9.94%
Annualized volatility	4.91%	19.35%
Sharpe (NAV-pct)	1.09	0.59
Sortino	1.57	
Max drawdown	-18.6%	-55.5%
Calmar	0.29	
Final NAV ($10M start)	$27,729,592	
Worst day	-3.93%	
% positive days	50.4%	
Sharpe 1996-2001	1.76	
Sharpe 2002-2007	1.45	
Sharpe 2008-2013	0.19	

Correlation of daily returns with the market: **-0.02**.

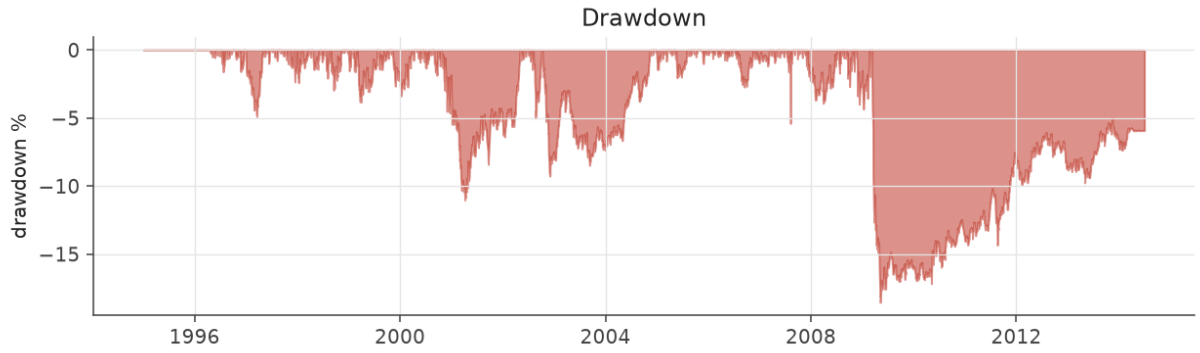
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 1.09

• best return among equity: `market_boost_max` (0.15); this one: 0.0538

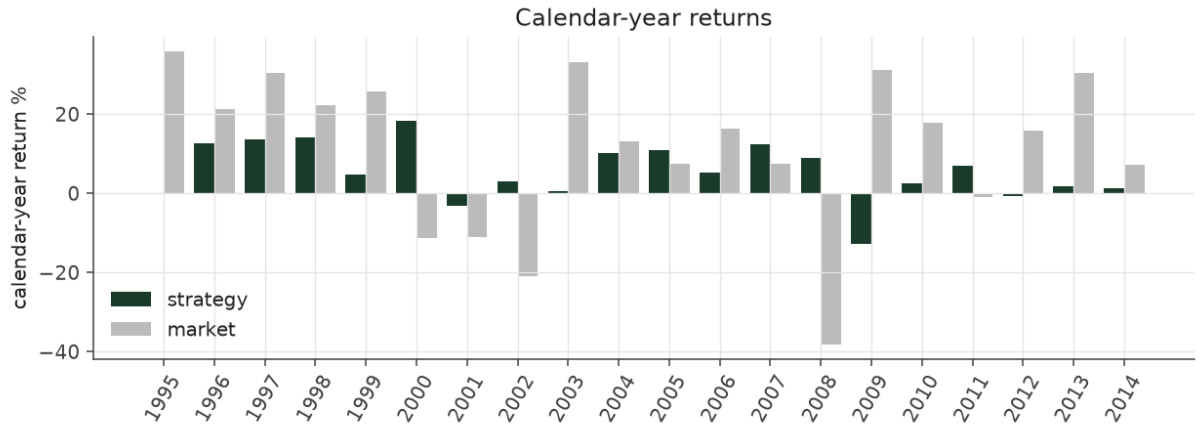
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.29



NAV (log scale)



Drawdown from running peak

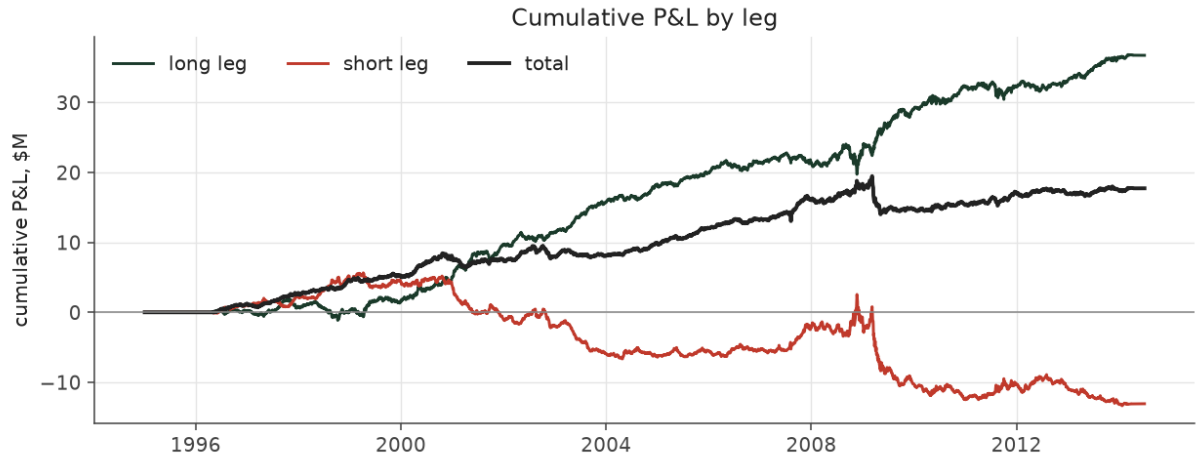


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	36.74	-13.08
P&L after costs ($M)	33.86	-16.13
of which common (average-event) component ($M)	23.41	-24.03
of which excess / PEAD component ($M)	13.34	10.96
excess minus costs ($M)	10.45	7.91
gross return per $ traded (bp)	252.72	-87.98
positions in the money (hit rate)	0.56	0.48
stand-alone Sharpe (leg only)	1.20	-0.33

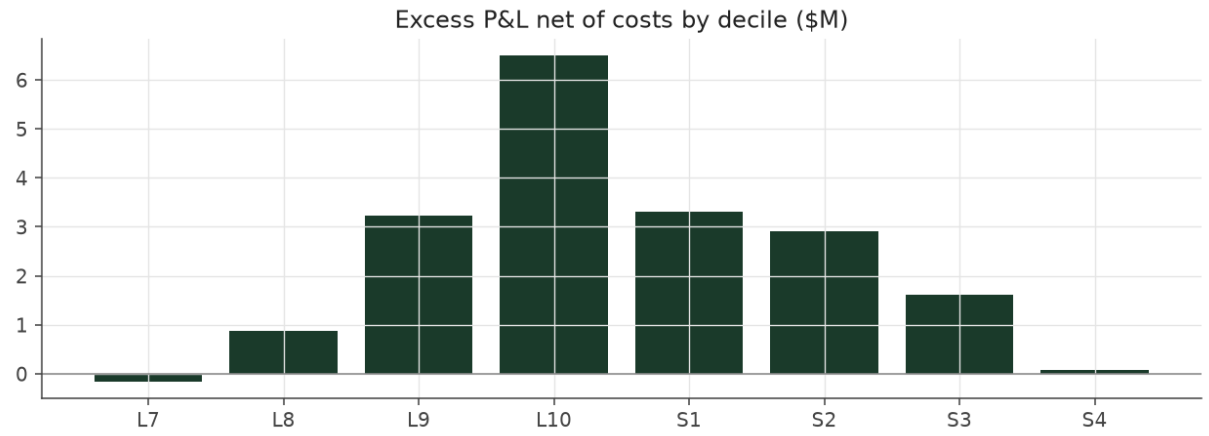
Daily P&L correlation between the two legs: **-0.80**.

• **The short leg loses money in absolute terms** (-13.1M before costs) while the long leg earns +36.7M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 23.4M / 24.0M. Stripping that common component, the surprise signal itself earned +13.3M on the long side and +11.0M on the short side.

• The two legs' daily P&L are -0.80 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.20 long / -0.33 short vs 1.09 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	5,756	95.09	1.46	0.03	0.19	-0.17	-0.9%	-17.41
long	8	15,259	315.71	6.61	1.51	0.63	0.88	4.8%	27.90
long	9	17,582	447.52	9.52	4.12	0.88	3.24	17.6%	72.33
long	10	19,546	595.59	16.27	7.68	1.18	6.50	35.4%	109.09
short	1	17,711	595.10	-6.77	4.58	1.27	3.31	18.0%	55.59
short	2	17,925	511.12	-5.46	3.94	1.03	2.91	15.8%	56.88
short	3	14,022	338.63	-3.40	2.29	0.68	1.62	8.8%	47.74
short	4	2,169	41.38	-0.50	0.15	0.08	0.08	0.4%	18.55



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,094	277.14	0.12	1.83	1.06	0.77	4.2%	27.79
2	21,037	558.41	6.62	8.55	1.56	6.98	38.0%	125.07
3	23,664	672.81	5.00	6.42	1.41	5.01	27.3%	74.49
4	25,541	712.39	3.70	4.61	1.11	3.49	19.0%	49.05
5	26,634	719.38	2.29	2.89	0.79	2.10	11.4%	29.18

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,373	201.26	2.28	3.67	1.01	2.67	14.5%	132.52
2	21,374	594.43	4.04	6.62	1.78	4.84	26.4%	81.40
3	23,903	716.16	5.43	7.18	1.43	5.75	31.3%	80.22
4	25,542	717.00	3.84	4.39	1.00	3.39	18.4%	47.22

5	26,774	711.11	2.13	2.42	0.71	1.70	9.3%	23.97

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	38,892	1,003.09	0.23	7.87	1.91	5.96	32.5%	59.44
41-60d	26,124	736.12	2.12	5.39	1.61	3.78	20.6%	51.39
61-90d	35,980	971.10	9.61	6.14	1.78	4.35	23.7%	44.84
>90d	8,974	229.82	5.76	4.90	0.64	4.26	23.2%	185.34

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,082	648.44	7.56	9.08	1.28	7.81	42.5%	120.38
2002-07	36,273	996.05	8.23	10.09	1.97	8.12	44.2%	81.55
2008-13	41,615	1,295.64	1.94	5.12	2.69	2.43	13.2%	18.75

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	19,612	276.98	0.79	1.35	0.58	0.77	4.2%	27.70
Chems	2,545	200.33	0.95	1.33	0.36	0.97	5.3%	48.46
Durbl	2,812	217.25	1.56	2.10	0.40	1.70	9.2%	78.15
Enrgy	6,250	246.54	3.05	3.55	0.47	3.08	16.8%	124.98
Hlth	10,936	280.00	1.24	1.86	0.59	1.27	6.9%	45.36
Manuf	11,449	262.67	1.40	2.01	0.55	1.46	7.9%	55.51
Money	15,175	240.41	1.92	2.55	0.55	2.00	10.9%	83.10
NoDur	4,887	242.43	1.08	1.49	0.47	1.02	5.6%	42.24
Other	16,350	265.41	1.89	2.56	0.59	1.97	10.7%	74.26
Shops	11,320	275.81	2.69	3.28	0.57	2.70	14.7%	98.05
Telcm	3,942	242.50	1.36	1.86	0.45	1.41	7.7%	58.27
Utils	4,692	189.81	-0.21	0.36	0.35	0.00	0.0%	0.20

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	36,794	943.20	1.67	4.44	1.72	2.73	14.9%	28.92
low vol	36,578	998.17	8.19	10.51	2.22	8.29	45.2%	83.09
mid vol	34,866	948.23	7.19	8.60	1.90	6.70	36.5%	70.69

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	5.38%	0.00%
0.5%	1.5	5.08%	0.30%
1.0%	3.0	4.76%	0.62%
2.0%	6.0	4.07%	1.31%
5.0%	15.0	1.26%	4.12%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.